NOTE **When not to use Bayesian optimization**

It's also important for us to know when the problem setting isn't appropriate and when not to use Bayesian optimization. As we have said, Bayesian optimization is useful when our limited resources prevent us from evaluate the objective function many times. If this is not the case, and evaluating the objective is cheap, we have no reason to be frugal with how we observe the objective function.

Here, if we can inspect the objective thoroughly across a dense grid, that will ensure that the global optimum is located. Otherwise, other strategies such as direct algorithms or evolutionary algorithms, which are algorithms that often excel at optimization when evaluations are cheap, may be used.

I hope this chapter was able to whet your appetite and get you excited for what's to come. In the next section, we summarize the key skills that you will be learning throughout the book.

1.3 What will you learn in this book?

This book gives you a deep understanding of the Gaussian process model and the Bayesian optimization task. You will learn how to implement a Bayesian optimization pipeline in Python using state-of-the-art tools and libraries. You will be exposed to a wide range of modeling and optimization strategies when approaching a Bayesian optimization task. By the end of the book, you will be able to:

- Implement high-performance Gaussian process models using GPyTorch, the premiere GP modeling tool in Python, visualize and evaluate their predictions, choose appropriate parameters for these models, and implement extensions such as variational Gaussian processes and Bayesian neural networks to scale to big data,
- Implement a wide range of Bayesian optimization policies using the state-of-the-art Bayesian optimization library BoTorch, which integrates nicely with GPyTorch, and inspect as well as understand their decision-making strategies,
- Approach different specialized settings such as batch, constrained, and multiobjective optimization using the Bayesian optimization framework, and
- Apply Bayesian optimization to a real-life task such as tuning the hyperparameters of a machine learning model.

Further, we will be using real-world examples and data in the exercises to consolidate what we learn in each chapter. In many different settings, we will run our algorithms on the same data set so that we can compare and analyze the different approaches taken.

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